

MM7020 Mathematical Communication

Scientific Project

An introduction to symplectic geometry

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1 Introduction

Take two vectors and compute its vector product. You probably thought of vectors in $\mathbb{R}^3$ but, can this be done in another space? Of which dimension? The truth is that a non-trivial vector product can only be defined in dimensions 3 and 7, and in the first case, it allows to define a *symplectic* structure on the smooth manifold $\mathbb{S}^2$, the two-dimensional sphere living in $\mathbb{R}^3$.

In this work we carry out an introduction to symplectic geometry. Roughly speaking, symplectic geometry deals with smooth manifolds equipped with a differential form satisfying certain properties. This branch of differential geometry was originally conceived to describe phase spaces in Hamiltonian mechanics in the 19th century. Hence, many of its applications are related to this field, especially flow problems in mechanical systems. A modern mathematical approach was performed in the mid 20th century, namely by H. Weyl, V. Arnold and A. Weinstein, who published one of the first compilations on the topic [Wei81].

We provide an introduction accessible to readers with basic knowledge in algebra and analysis. Starting by the definition of smooth manifold, in Section 2 we review some of the fundamental concepts in differential geometry, including smooth maps, the tangent and cotangent bundles and tensor fields. Next, in Section 3, we introduce the more technical notion of exterior product of vector spaces, that allows us to define differential forms of any order on a manifold and the exterior derivative. These sections cover basic results on differential manifolds and algebraic structures. We include them in order to have a comprehensive introduction to symplectic manifolds, but do not show the proofs of the results stated. The details can be found in general texts such as [BC70], [DF03], [Lan06] or [War83].

With all these tools, in Section 4 we introduce symplectic manifolds and include three fundamental results. First, we show that symplectic manifolds have even dimension (Corollary 4.4). Then, we formulate Darboux theorem, which states that every symplectic manifold is locally equivalent to $\mathbb{R}^{2n}$ (Theorem 4.7). Finally, we equip the cotangent bundle of any manifold with a symplectic structure (Theorem 4.10). The main reference followed in this part is [Can06].

2 Smooth manifolds

There exist different definitions of manifolds, some of them starting from just a set without any structure, constructing the so-called *abstract* manifolds. However, given that most of the interesting properties of manifolds require some topological structure, we define them directly on Hausdorff spaces.

Thus, let us start considering a Hausdorff topological space M , together with an open cover $\{U_i\}_{i \in I}$ of M . A collection of pairs (U_i, φ_i) with $i \in J \subseteq I$ is called an **atlas** (of class $\mathcal{C}^p$) on M if it satisfies the following conditions:

1. Each φ_i is a homeomorphism of U_i onto an open subset $\varphi_i(U_i) \subseteq \mathbb{R}^n$ and for any $i, j \in J$, $\varphi_i(U_i \cap U_j)$ is open in $\mathbb{R}^n$.
2. The map

$$\varphi_j \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \longrightarrow \varphi_j(U_i \cap U_j)$$

is a $\mathcal{C}^p$ -diffeomorphism¹ for each pair of indices $i, j \in J$.

Each pair (U_i, φ_i) is then called a **chart** of the atlas.

Note that the concept of atlas can be defined more generally replacing $\mathbb{R}^n$ with an arbitrary vector space E_i , which need not be the same for every chart. We will however stick to the real case for simplicity. In addition, throughout the text we will assume the class to be $\mathcal{C}^\infty$.

Given an open subset $U \subseteq M$ and a homeomorphism $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$, we say that (U, φ) is compatible with the atlas $\{(U_i, \varphi_i)\}_{i \in J}$ if each map $\varphi_i \varphi^{-1}$ is a diffeomorphism. Consequently, we say two atlas to be compatible if every chart of one is compatible with the other. It can be easily verified that the notion of compatibility induces an equivalence relation.

Definition 2.1. With the notation above, an n -dimensional **smooth manifold** (or differential manifold) is an equivalence class of atlases on M .

Equivalently, when we assign a class of atlases to such a space M , we say that M is equipped with a differentiable structure. In the following we may indistinctly use the terms manifold, differentiable manifold and smooth manifold.

2.1 Mappings on manifolds

In this section we discuss the basic notions of mappings between manifolds and set the notation that we use in later arguments. Since we are working on a topological space and the charts have been defined as homeomorphisms, continuous maps between manifolds are determined by the topology of M . Furthermore, the connection of manifolds with $\mathbb{R}^n$ allows us to extend the concept of differentiability in these spaces to manifolds.

Throughout the text, a map $f : M \rightarrow \mathbb{R}$ is said to be a smooth function at $x \in M$ if $f \circ \varphi^{-1}$ is differentiable at $\varphi(x)$ for every chart (U, φ) such that $x \in U$. If f is smooth at every $x \in M$, then it is called a **smooth** (or differentiable) **function**. Note that with this definition the charts are smooth functions. We denote the set of all smooth functions on a manifold M by $\mathfrak{F}(M)$.

In a similar fashion, given two smooth manifolds M and N , a map $\psi : M \rightarrow N$ is said to be a smooth map at $x \in M$ if $\varphi' \circ \psi \circ \varphi^{-1} : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is differentiable at $\varphi(x)$ for every chart (U, φ) on M with $x \in U$ and every chart (U', φ') on N with $\psi(x) \in U'$. If ψ is smooth at every $x \in M$, then it is called a **smooth map** between manifolds. In addition, if the smooth map ψ is bijective and its inverse is also smooth, then it is called a **diffeomorphism** between manifolds.

2.2 Tangent bundle

We have seen that manifolds are n -dimensional generalisations of curves and surfaces. In this section, we generalise the idea of the tangent line of a curve or the tangent plane of a surface, by defining in general the tangent space of a manifold at a certain point.

We recall that given a curve $\gamma = \gamma(t)$ in $\mathbb{R}^2$, the tangent line at the point $\gamma(t_0)$ is the one-dimensional affine space in $\mathbb{R}^2$ containing $\gamma(t_0)$ and with direction $\gamma'(t_0)$. We observe that this does not depend on the parametrisation of the curve, and thus is well-defined. For a simple surface $f(u, v)$ in $\mathbb{R}^3$, the tangent plane at a point $f(u_0, v_0)$ is the two-dimensional affine

¹Some texts define diffeomorphisms in terms of manifolds, but here we call diffeomorphism (of class $\mathcal{C}^p$) simply to a bijective map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that both φ and its inverse φ^{-1} are of differentiability class $\mathcal{C}^p$.

plane in $\mathbb{R}^3$ containing $f(u_0, v_0)$ and all the tangent vectors at $\gamma'(0)$ of curves $\gamma(t)$ such that $\gamma(0) = f(u_0, v_0)$. This is the idea in the general n -dimensional case.

Let M be an n -dimensional smooth manifold and $x \in M$. Let $\varphi : U \subseteq M \rightarrow \mathbb{R}^n$ be a chart containing x . Without loss of generality, we can assume that $\varphi(x) = \mathbf{0}$, since this can be achieved by composition with a translation. Now we consider a curve $\gamma : (-\varepsilon, \varepsilon) \subseteq \mathbb{R} \rightarrow M$ such that $\text{im} \gamma \subseteq U$ and $\gamma(0) = x$. Then, the composition $\varphi \circ \gamma$ is a curve on $\mathbb{R}^n$. It can be shown that the tangent vector of this curve at $t = 0$, that we denote by $(\varphi \circ \gamma)'(0)$, does not depend on the chart considered. This allows us to define the following equivalence relation: given two curves γ_1, γ_2 on M such that $\gamma_1(0) = \gamma_2(0) = x$, they are equivalent if

$$(\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0)$$

as vectors in $\mathbb{R}^n$.

Definition 2.2. Let M be a manifold and $x \in M$, and consider the equivalence relation above defined on the set of curves that contain x . The equivalence classes of the relation are called **tangent vectors** of M at x , and the quotient set is called the **tangent space** of M at x . It is denoted by $T_x M$.

Next, given a chart $\varphi : U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ such that $\varphi(x) = \mathbf{0}$, we consider for each $i \in \{1, \dots, n\}$ the curve

$$\gamma_i(t) := \varphi^{-1}((0, \dots, t, \dots, 0)) = \varphi^{-1}(\mathbf{0} + te_i),$$

also called **coordinate curve** at the i -th coordinate. The representative tangent vector of this curve in the tangent space is denoted by $\left(\frac{\partial}{\partial x^i}\right)_x$. This notation has its origin in the action of vector fields on the set of functions $\mathfrak{F}(M)$, that we consider further in this section. The following result shows the importance of these special tangent vectors in our reasoning.

Theorem 2.3. *Given an n -dimensional manifold M and any $x \in M$, the tangent space $T_x M$ is a real vector space of dimension n . Moreover, the following is a basis of $T_x M$:*

$$\left\{ \left(\frac{\partial}{\partial x^1} \right)_x, \dots, \left(\frac{\partial}{\partial x^n} \right)_x \right\}.$$

The (disjoint) union of the tangent spaces at all points of a manifold,

$$TM := \bigcup_{x \in M} T_x M,$$

is called the **tangent bundle** of the manifold M .

Example. Consider the unit circle $\mathbb{S}^1$ in $\mathbb{R}^2$, which can easily be verified to be a one-dimensional smooth manifold. The tangent space $T_x \mathbb{S}^1$ at a certain point $x \in \mathbb{S}^1$ can be interpreted as a line tangent to the circle in x . The tangent bundle $T\mathbb{S}^1$ is therefore the disjoint union of all such lines. In particular, we can represent them being disjoint in $\mathbb{R}^3$, perpendicular to the plane in which $\mathbb{S}^1$ is contained, so that $T\mathbb{S}^1$ is diffeomorphic to the cylinder $\mathbb{S}^1 \times \mathbb{R}$.

We can understand the tangent bundle as the set $TM = \{(x, v) : x \in M, v \in T_x M\}$. In other words, the tangent bundle can be locally seen as the product of a chart domain U and an n -dimensional vector space at each point $x \in U$, as was the case in the previous example. The following general result shows that this idea can be extended to the whole manifold. Despite being an intuitive statement, the proof is not straightforward.

Proposition 2.4. *The tangent bundle TM of an n -dimensional smooth manifold M is a smooth manifold of dimension $2n$.*

In this way, the natural projection $\rho : TM \rightarrow M$ such that $\rho(x, v) = x$ is a smooth map between manifolds. The smooth assignment of a vector $v \in T_x M$ to each point $x \in M$ is called a vector field. In other words, a vector field is a smooth section of the projection $\rho : TM \rightarrow M$. However, in the case of bundles it is common to refer to these maps simply as sections of the bundle, assuming that the involved map is the projection onto the original manifold. The formal definition follows.

Definition 2.5. A **vector field** is a smooth section of the tangent bundle.

As we mentioned before, vector fields can also be understood as maps $\mathfrak{F}(M) \rightarrow \mathfrak{F}(M)$. Indeed, a vector field X can be described in local coordinates² by $X = \frac{\partial}{\partial x_i} X^i$, with X^i smooth functions on the chart domain. Then, for any function $f : M \rightarrow \mathbb{R}$, we define

$$X(f) := \frac{\partial f}{\partial x_i} X^i,$$

where the partial derivatives $\frac{\partial f}{\partial x_i}$ are the directional derivatives of f along the coordinate curves, which are well-defined because f is smooth and so is any chart. Additionally, it can be shown that this definition does not depend on the selected chart. Thus $X(f)$ is again a smooth function on M , and the map is well-defined.

2.3 Cotangent bundle

The next step is to consider linear maps from the tangent space into $\mathbb{R}$. In general, given a vector space V over a field $\mathbb{K}$, a scalar product $\langle \cdot, \cdot \rangle$ on V and a fixed $v \in V$, the map $w \mapsto \langle v, w \rangle$ is a linear map $V \rightarrow \mathbb{K}$. It is usually denoted by v^* . In particular, we can define linear maps $e^i := e_i^*$ such that $e^i(e_j) = \delta_j^i$ for each e_j in the canonical basis.

Theorem 2.6. *Let V be an n -dimensional vector space over a field $\mathbb{K}$. The set V^* of all linear transformations $V \rightarrow \mathbb{K}$ is again an n -dimensional vector space over $\mathbb{K}$, and a basis is given by the set $\{e^1, \dots, e^n\}$.*

The vector space V^* is called the **dual vector space** of V . The basis $\{e^1, \dots, e^n\}$ is the **dual basis** of the canonical basis $\{e_1, \dots, e_n\}$ of V .

Definition 2.7. Let M be a smooth manifold and $x \in M$. The dual vector space of the tangent space $T_x M$ is called the **cotangent space** of M at x , and it is denoted by $T_x^* M$.

We denote by dx^i the dual element of the vector $\frac{\partial}{\partial x^i} \in T_x M$. In this way, the set

$$\{dx^1, \dots, dx^n\},$$

is a basis of $T_x^* M$ dual of the one in Theorem 2.3. Just like before, we consider the union of the cotangent spaces of points in M , which will allow us to define differential forms. The set

$$T^* M := \bigcup_{x \in M} T_x^* M,$$

is called the **cotangent bundle** of the manifold M . Similarly, we can interpret the cotangent bundle as $T^* M = \{(x, \xi) : x \in M, \xi \in T_x^* M\}$.

²For the sake of brevity, it is common in differential geometry to use the Einstein summation convention, introduced by the German physicist in his works on General Relativity. In this notation, an index appearing twice (first as a lower index and then as an upper index) denotes a sum over all the values of the index. We use this notation over the whole work. In particular, in linear combinations the coefficients need not precede the vectors, as normally happens. One can always translate the expressions to the usual notation if Einstein's convention is confusing.

Proposition 2.8. *The cotangent bundle T^*M of an n -dimensional smooth manifold M is a smooth manifold of dimension $2n$.*

Once again we have a natural projection $\pi : T^*M \rightarrow M$ such that $\pi(x, \xi) = x$.

Definition 2.9. A **(differential) 1-form** is a smooth section of the cotangent bundle.

In other words, a 1-form α maps each point $x \in M$ into a linear map $\alpha_x : T_x M \rightarrow \mathbb{R}$, which itself maps every vector $v \in T_x M$ into a real number $\alpha_x(v)$. These linear maps are also known as $(0,1)$ -tensors, and hence a 1-form is a $(0,1)$ -tensor field. More generally, we can consider multilinear maps $(T_x M)^k \rightarrow \mathbb{R}$, defined on k -tuples of vectors in $T_x M$. Such maps are called $(0,k)$ -tensors, and the smooth assignment of a $(0,k)$ -tensor at each $x \in M$ is known as $(0,k)$ -tensor field. The differential k -forms, which generalise 1-forms, are the *alternated* $(0,k)$ -tensor fields. We define them precisely in the following section.

Remark 2.10. Differential 1-forms, and in general k -forms, map each point $x \in M$ to a multilinear transformation α_x defined on $(T_x M)^k$, usually called multilinear form. However, some sources use the same term to refer both to differential form and to the local multilinear form α_x , which might lead to confusion. Therefore, in this work we call differential form the map defined on M , and use other terms to refer to the induced map at x .

3 Exterior product and differential forms

The cotangent bundle has emerged as the necessary structure to define 1-forms, since it contains all the linear maps $T_x M \rightarrow \mathbb{R}$ at each point $x \in M$. Now we wish to construct an equivalent structure containing exactly all the alternated maps $(T_x M)^k \rightarrow \mathbb{R}$. The starting point for this description is the tensor product of vector spaces.

Definition 3.1. Let V, W be two vector spaces. The **tensor product** $V \otimes W$ is a vector space such that:

1. There exists a bilinear map $\theta : V \times W \rightarrow V \otimes W$.
2. For any bilinear map $\phi : V \times W \rightarrow Z$, there exists a unique linear map $\tilde{\phi} : V \otimes W$ such that the following diagram commutes

$$\begin{array}{ccc} V \times W & \xrightarrow{\theta} & V \otimes W \\ & \searrow \phi & \downarrow \tilde{\phi} \\ & & Z \end{array} .$$

We speak of *the* tensor product $V \otimes W$ as it is uniquely determined up to isomorphism. Its elements are normally denoted by $v \otimes w := \theta(v, w)$ for each pair $(v, w) \in V \times W$.

Theorem 3.2. *Let $\{e_1, \dots, e_n\}$ and $\{f_1, \dots, f_m\}$ be bases of two vector spaces V and W of dimensions n and m , respectively. Then, the tensor product $V \otimes W$ has dimension $n \cdot m$, and a basis is given by the set*

$$\{e_i \otimes f_j : i \in \{1, \dots, n\}, j \in \{1, \dots, m\}\}.$$

Intuitively, the tensor product can be seen as the vector space whose basis serves as a basis for bilinear maps defined on $V \times W$. Hence, the space of bilinear maps $V \times W \rightarrow \mathbb{K}$ is isomorphic to the space of linear maps $V \otimes W \rightarrow \mathbb{K}$. The latter is precisely the dual vector space $(V \otimes W)^*$, which turns out to be isomorphic to $V^* \otimes W^*$. More generally, we can inductively consider the

tensor product $\bigotimes_{i=1}^k V_i^*$. Thus, when $V_i = V$ for all $i \in \{1, \dots, k\}$, we have an isomorphism between the space of multilinear maps $\prod_{i=1}^k V \rightarrow \mathbb{K}$ and the tensor product $\bigotimes^k V^* := \bigotimes_{i=1}^k V^*$, whose elements are called $(0, k)$ -tensors. For the sake of completeness, we observe that the notion of (ℓ, k) -tensor arises when considering the elements of the space

$$\left(\bigotimes^{\ell} V \right) \otimes \left(\bigotimes^k V^* \right).$$

However, in this work we restrict ourselves to the tensors with $\ell = 0$, called covariant tensors.

3.1 Exterior product

So far we have not required the multilinear transformations in $\bigotimes^k V^*$ to be alternated. For that, we define the following equivalence relation on $\bigotimes^k V^*$:

$$(v_1^* \otimes \dots \otimes v_k^*) \sim (w_1^* \otimes \dots \otimes w_k^*) \iff v_i^* - w_i^* = v_j^* - w_j^* \text{ for some } i \neq j.$$

Remark 3.3. This can be done more precisely by noting that the tensor product has actually the structure of a module and considering its quotient modulo the ring generated by elements $v^* \otimes v^*$, but we stick to the intuitive idea here. Likewise, the derivation is valid for any vector space, but we work with the dual of V because we will apply these notions on the cotangent spaces.

The resulting quotient set is again a vector space over $\mathbb{K}$, known as the k -th **exterior power** of V^* . It is denoted by $\bigwedge^k V^* := \bigwedge_{i=1}^k V^*$. The equivalence class of $v_1^* \otimes \dots \otimes v_k^*$ in $\bigwedge^k V^*$ is denoted by $v_1^* \wedge \dots \wedge v_k^*$. The basis elements of $\bigotimes^k V^*$ which are not related by $\sim$ form a basis of $\bigwedge^k V^*$, that is the set

$$\{e_{\ell_1}^* \wedge \dots \wedge e_{\ell_k}^* : \ell_i \neq \ell_j \text{ for } i, j \in \{1, \dots, k\}\}.$$

It follows that the dimension of the k -th exterior power of an n -dimensional vector space is $\binom{n}{k}$. Note that it is trivial for $k > n$.

Furthermore, we can define $\wedge$ as a binary operator acting on elements of different exterior powers of V^* . Given $v^* = v_1^* \wedge \dots \wedge v_k^* \in \bigwedge^k V^*$ and $w^* = w_1^* \wedge \dots \wedge w_m^* \in \bigwedge^m V^*$, we define

$$v^* \wedge w^* := v_1^* \wedge \dots \wedge v_k^* \wedge w_1^* \wedge \dots \wedge w_m^* \in \bigwedge^{k+m} V^*.$$

This operation is called the **exterior product** of the vectors v^* and w^* . As a direct consequence of the equivalence relation considered, the exterior product of two vectors is **antisymmetric**, i.e. $v^* \wedge w^* = -w^* \wedge v^*$ for all $v^*, w^* \in V^*$. In this way, taking $\mathbb{K}$ as the 0-th exterior power of V^* , the set

$$\bigwedge V^* = \bigoplus_{k=0}^{\infty} \bigwedge^k V^* = \bigoplus_{k=0}^n \bigwedge^k V^*$$

is a $\mathbb{K}$ -algebra, known as the **exterior algebra** (or also **Grassmann algebra**) of V^* .

3.2 Exterior derivative in manifolds

We consider again a smooth manifold M and a point $x \in M$. For $k \in \mathbb{N}$, let $\bigwedge^k T_x^* M$ be the k -th exterior power of the cotangent space at x .

Definition 3.4. A **differential k -form** is a smooth map

$$\begin{aligned} \alpha : M &\longrightarrow \bigcup_{y \in M} \left(\bigwedge^k T_y^* M \right) \\ x &\longmapsto \alpha_x \end{aligned}$$

such that $\alpha_x \in \bigwedge^k T_x^* M$ for all $x \in M$.

We denote by $\Omega^k(M)$ the set of all k -forms on the manifold M , and by $\Omega(M) = \bigcup_{k \in \mathbb{N}} \Omega^k(M)$ the set of differential forms of any order defined on M .

In other words, similarly as we did with 1-forms, we can define k -forms as smooth sections of the bundle whose fibers are the exterior powers $\bigwedge^k T_x^* M$ at each $x \in M$. Since the underlying field of the manifolds considered is $\mathbb{R}$, 0-forms smoothly assign to each $x \in M$ a real number, i.e. they are in bijection with the set of functions $\mathfrak{F}(M)$. This bijection is the starting point for defining the *exterior derivative*, which generalizes on differential forms of any order the concept of differential of a function. Before giving the proper definition, we note that given two differential forms $\alpha, \beta \in \Omega(M)$, we can define their exterior product pointwise in $\bigwedge T_x^* M$ for each $x \in M$ as

$$(\alpha \wedge \beta)_x := \alpha_x \wedge \beta_x.$$

Definition 3.5. Let M be a smooth manifold, and let $d : \Omega(M) \rightarrow \Omega(M)$ be an $\mathbb{R}$ -linear map that maps any form $\alpha \in \Omega^k(M)$ to $d\alpha \in \Omega^{k+1}(M)$, for all $k \in \mathbb{N}$. The map d is called the **exterior derivative** on M if it satisfies the following properties:

1. For any function $f \in \Omega^0(M)$ and vector field X on M , $df(X) = X(f)$. This map is usually known as the differential of the function f .
2. For any function $f \in \Omega^0(M)$, $d^2 f = 0$.
3. For any two forms $\alpha \in \Omega^k(M)$, $\beta \in \Omega^\ell(M)$, $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^k \alpha \wedge d\beta$.

Just like when we defined the tensor product, we speak of *the* exterior derivative since it is the unique such map satisfying these properties. In some occasions, the definition includes the second property for *any* k -form. However, that can be derived from the stated properties.

The fact that d transforms k -forms into $(k+1)$ -forms allows to study differential forms using the machinery of algebraic topology, by means of the so-called **de Rham cohomology**. The following concepts arise naturally from that construction.

Definition 3.6. Let $k \in \mathbb{N}$.

- A differential form $\alpha \in \Omega^k(M)$ is called **closed** if $d\alpha = 0$.
- A differential form $\alpha \in \Omega^{k+1}(M)$ is called **exact** if $\alpha = d\beta$ for some $\beta \in \Omega^k(M)$.

Since $d^2 = 0$, every exact differential form is closed. However, the converse is not true in general. A counterexample in a non-contractible space is the exterior derivative $d\theta$ in $\mathbb{R}^n \setminus \{\mathbf{0}\}$. However, Poincaré's lemma states that it holds if the domain is contractible, so for instance in all the $\mathbb{R}^n$ [War83].

3.3 Differential and pullback

In the previous section we extended the notion of differential of a function (i.e. a 0-form) to k -forms in general. Now we interpret the differential as a linear approximation of the function around a given point. In this way, another generalisation of the differential consists in extending this idea to smooth maps between manifolds.

Definition 3.7. Let M, N be smooth manifolds and $\psi : M \rightarrow N$ a smooth map between them. The **differential** of ψ at a point $x \in M$ is defined as the linear map $(\psi_*)_x : T_x M \rightarrow T_{\psi(x)} N$ that sends every vector $v = \gamma'(0) \in T_x M$ to $(\psi_*)_x(v) = (\psi \circ \gamma)'(0) \in T_{\psi(x)} N$.

Remark 3.8. The differential defines globally a smooth map $\psi_* : TM \rightarrow TN$ between the tangent bundles of the manifolds. The image of a vector by this map is called the **pushforward** of the vector by ψ_* . In turn, the map ψ_* induces a *bundle* map from TM into the so-called *pullback bundle* induced by ψ , commonly known as tangent map. However, its formal definition is beyond the scope of this work, as we will only use the local map between tangent spaces in the text.

Note that if $N = \mathbb{R}$, then the map $(\psi_*)_x$ equals the (usual) differential of ψ at x , i.e. the map $(d\psi)_x$ obtained from the exterior derivative of the 0-form ψ . In fact, the differential of a smooth map is sometimes denoted also by $d\psi$, but we avoid this notation here for the sake of clarity.

The differential is a useful tool to define differential forms on M starting from a differential form on N , that is, to pull forms on N back to M .

Definition 3.9. Let $\psi : M \rightarrow N$ be a smooth map and $\alpha \in \Omega^k(N)$ a k -form on N . Then the **pullback** of the form α induced by the map ψ is the k -form $\psi^*\alpha \in \Omega^k(M)$ such that

$$(\psi^*\alpha)_x(v_1, \dots, v_k) = \alpha_{\psi(x)}((\psi_*)_x(v_1), \dots, (\psi_*)_x(v_k)),$$

for all $x \in M$ and $v_1, \dots, v_k \in T_x M$. It is also said that ψ pulls α back to $\psi^*\alpha$.

Thus, we can understand the pullback of differential forms as a map $\psi^* : \Omega(N) \rightarrow \Omega(M)$. This transformation is remarkably useful since it commutes with the exterior product and the exterior derivative. In other words, given differential forms $\alpha, \beta \in \Omega(N)$,

$$\psi^*(\alpha \wedge \beta) = (\psi^*\alpha) \wedge (\psi^*\beta), \quad \text{and} \quad \psi^*(d\alpha) = d(\psi^*\alpha).$$

4 Symplectic manifolds

In this last section, we use the concepts introduced in the sections above to define the notion of symplectic manifold and state some basic properties. For that, we need first to define a symplectic map on a vector space, which is a non-degenerate skew-symmetric bilinear form.

4.1 Skew-symmetric bilinear forms

Let V be an n -dimensional vector space over a field $\mathbb{K}$.

Definition 4.1. A bilinear map $\phi : V \times V \rightarrow \mathbb{K}$ is called **skew-symmetric** if $\phi(u, v) = -\phi(v, u)$ for all $u, v \in V$.

Theorem 4.2. Let $\phi : V \times V \rightarrow \mathbb{K}$ be a skew-symmetric bilinear map. Then there is a basis $\{u_1, \dots, u_k, e_1, \dots, e_m, f_1, \dots, f_m\}$ such that

- (1) $\phi(u_i, v) = 0$ for all $v \in V$,
- (2) $\phi(e_i, e_j) = \phi(f_i, f_j) = 0$ for all $i, j \in \{1, \dots, m\}$,
- (3) $\phi(e_i, f_j) = \delta_{ij}$ for all $i, j \in \{1, \dots, m\}$.

Proof. Consider $U := \{u \in V : \phi(u, v) = 0 \text{ for all } v \in V\}$, with basis $u_1, \dots, u_k$. We can choose a complementary space W to U in V , that is, $V = U \oplus W$. Given a nonzero $e_1 \in W$, there exists $f_1 \in W$ such that $\phi(e_1, f_1) \neq 0$, so without loss of generality we can assume $\phi(e_1, f_1) = 1$. Denote W_1 the space spanned by e_1 and f_1 , and $W_1^\phi := \{w \in W : \phi(w, v) = 0 \text{ for all } v \in W_1\}$. Let's see that $W = W_1 \oplus W_1^\phi$:

- $W_1 \cap W_1^\phi = \{0\}$, since given $v = ae_1 + bf_1 \in W_1 \cap W_1^\phi$, the condition $\phi(v, e_1) = 0 = \phi(v, f_1)$ implies $a = 0 = b$, so $v = 0$.
- Given $w \in W$ with $\phi(w, e_1) = c$ and $\phi(w, f_1) = d$, we can write $w = (-cf_1 + de_1) + (w + cf_1 - de_1)$, the first term belonging to W_1 and the second to W_1^ϕ .

We can repeat the process with a nonzero element $e_2 \in W_1^\phi$, constructing a new space W_2 spanned by e_2 and f_2 such that $\phi(e_2, f_2) = 1$. Since $\dim V < \infty$, after finitely many steps we obtain $V = U \oplus W_1 \oplus W_2 \oplus \dots \oplus W_n$, where all addends are orthogonal with respect to ϕ , and where W_i has basis e_i, f_i with $\phi(e_i, f_i) = 1$. $\square$

The set U defined in the proof is the set of vectors orthogonal to all V with respect to ϕ . Note that $\dim U = k$. Next, we consider the dual space V^* of linear transformations $V \rightarrow \mathbb{K}$, and define the linear map $\tilde{\phi} : V \rightarrow V^*$ such that $\tilde{\phi}(v)(u) = \phi(v, u)$ for each $v \in V$.

Definition 4.3. A skew-symmetric bilinear map $\phi : V \times V \rightarrow \mathbb{K}$ is called **symplectic** if the map $\tilde{\phi}$ is bijective, i.e. $U = \{0\}$. The pair (V, ϕ) is then called a **symplectic vector space**.

Corollary 4.4. *Every symplectic vector space has even dimension.*

Proof. Following the statement of Theorem 4.2, the subspace U has dimension k , and we can write $n = \dim V = k + 2m$. In the case of a symplectic vector space, we know that $U = \{0\}$, i.e. $k = 0$, and therefore $n = 2m$ is even. $\square$

4.2 Symplectic manifolds

All the previous background allows us now to give the complete definition of a symplectic manifold, which is no more than a manifold equipped with a closed, symplectic 2-form. We switch to denoting by p the points in a symplectic manifold, to differentiate with those of a general manifold, which we still call x .

Definition 4.5. Let M be a smooth manifold. A differential 2-form $\omega \in \Omega^2(M)$ is said to be **symplectic** if ω is closed and ω_p is a symplectic bilinear map for all $p \in M$. The pair (M, ω) is then called a **symplectic manifold**.

A first property of symplectic manifolds follows directly from Corollary 4.4: a symplectic manifold always has **even dimension**. Let us present some specific examples.

Examples.

1. The canonical two-dimensional real manifold, $\mathbb{R}^2$, admits a symplectic structure, just as $\mathbb{R}^{2n}$ for $n \geq 1$. We consider coordinates $x_1, \dots, x_n, y_1, \dots, y_n$, and define the form

$$\omega_0 = \sum_{i=1}^n dx^i \wedge dy^i,$$

which can easily be checked to be symplectic.

2. As we commented in the introduction, the sphere $\mathbb{S}^2 = \{p \in \mathbb{R}^3 \mid \|p\| = 1\}$ is a symplectic manifold, together with the form given by

$$\omega_p(u, v) = \langle p, u \times v \rangle \quad \text{for } u, v \in T_p\mathbb{S}^2.$$

This is the only symplectic sphere. Although the vector product can also be performed in $\mathbb{S}^6$, homological arguments show that a symplectic structure is not possible in any other dimension.

We wonder now how different symplectic structures on manifolds are related to each other. The concept of pullback defined in Section 3.3 allows us to compare symplectic forms in the following way.

Definition 4.6. Let (M_1, ω_1) and (M_2, ω_2) be $2n$ -dimensional symplectic manifolds, and let $\psi : M_1 \rightarrow M_2$ be a diffeomorphism. The map ψ is said to be a **symplectomorphism** if ψ pulls ω_2 back to ω_1 , i.e. if $\psi^*\omega_2 = \omega_1$.

The natural question that arises after this definition is how to classify symplectic manifolds up to symplectomorphism. Darboux theorem gives an answer to this. Just as every smooth manifold is locally diffeomorphic to $\mathbb{R}^n$, symplectic manifolds are locally symplectomorphic to $(\mathbb{R}^{2n}, \omega_0)$, no matter the symplectic structure chosen.

Theorem 4.7 (Darboux theorem). *Let (M, ω) be a $2n$ -symplectic manifold and $p \in M$. Then there is a chart (U, φ) with $\varphi(p) = \mathbf{0}$ such that*

$$\omega = \sum_{i=1}^n dx^i \wedge dy^i \quad \text{on } U.$$

We omit the details of the proof, which consists of defining a symplectic basis as in Theorem 4.2 and checking that the form defined this way in local coordinates coincides with ω on the whole U . It can be consulted in [Can06] (Theorem 8.1).

In the final part of the text, we give an example of construction of symplectic manifolds. In particular, we show that the cotangent bundle T^*M of every manifold M admits a symplectic structure, by constructing the so-called canonical symplectic 2-form. We remind that the dimension of T^*M is always twice the dimension of M , and hence even. Following the notation introduced in Section 2.3, the points of the cotangent bundle are pairs $p = (x, \xi)$ with $x \in M$ and $\xi \in T_x^*M$. We consider the canonical projection $\pi : T^*M \rightarrow M$ and its differential $(\pi_*)_p : T_p(T^*M) \rightarrow T_xM$ at each point $p \in T^*M$. With that, we define the **tautological form** $\alpha \in \Omega^1(M)$ by

$$\begin{aligned} \alpha_p : T_p(T^*M) &\longrightarrow \mathbb{R} \\ v &\longmapsto \xi((\pi_*)_p(v)) \quad , \end{aligned}$$

for each $p = (x, \xi) \in T^*M$.

Proposition 4.8. *Given a chart $\phi : U \rightarrow \mathbb{R}^n$ of M with associated cotangent coordinates $(x_i, \xi_i)_{i=1}^n$, the coordinate expression of the tautological form is given by*

$$\alpha = \xi_i dx^i.$$

Proof. Let $v = \frac{\partial}{\partial p^i} v^i$. Then

$$\alpha_p(v) = \xi((\pi_*)_p(v)) = \xi\left(\frac{\partial}{\partial x^i} v^i\right) = \xi\left(\frac{\partial}{\partial x^i}\right) v^i = \xi_i v^i = (\xi_i dx^i)(v).$$

□

Note that the representation does not depend on the chart.

The **canonical symplectic 2-form** is now defined by means of the exterior derivative as

$$\omega = -d\alpha.$$

Theorem 4.9. *The 2-form ω is a symplectic form.*

Proof. We need to check that ω is closed and ω_p is a symplectic bilinear map for all $p \in T^*M$. The first statement follows from the fact that $\omega = d(-\alpha)$ is an exact form.

For the second one, we consider $p \in T^*M$ and a chart domain U containing p . We can express α in local coordinates using Proposition 4.8. If we note that $\xi_i dx^i = \xi_i \wedge dx^i$ and apply Definition 3.5 (3), we have $\omega = -d\alpha = d\xi_i \wedge dx^i + \xi_i d^2x^i$. Using now (1) in Definition 3.5, since the coordinates ξ_i are smooth functions, we get $d\xi_i = d\xi^i$. Furthermore $d^2x^i = 0$. Hence, given that the exterior product is antisymmetric, we obtain the coordinate expression

$$\omega = \sum_{i=1}^n dx^i \wedge d\xi^i.$$

This representation is again independent of the chart selected, as the one of α was. Now let $\{e_j, f_j\}_{j=1}^n$ be a basis of $T_p(T^*M)$. We note that

$$\omega_p(e_j, f_j) = \left(\sum_{i=1}^n dx^i \wedge d\xi^i \right) (e_j, f_j) = \sum_{i=1}^n \delta_j^i \delta_k^i = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{otherwise,} \end{cases}$$

while $\omega_p(e_j, e_k) = \omega_p(f_j, f_k) = 0$ and $\omega_p(f_j, e_j) = -\omega_p(e_j, f_j)$ because of the antisymmetry of the exterior product. Hence, the bilinear map ω_p is a skew-symmetric transformation determined by a matrix of the form

$$\begin{pmatrix} 0 & -\text{Id} \\ \text{Id} & 0 \end{pmatrix},$$

whose determinant is ± 1 , and therefore is symplectic. $\square$

The final result of this work follows immediately.

Theorem 4.10. *The cotangent bundle of a manifold equipped with the canonical symplectic form is a symplectic manifold.*

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